Problem 2

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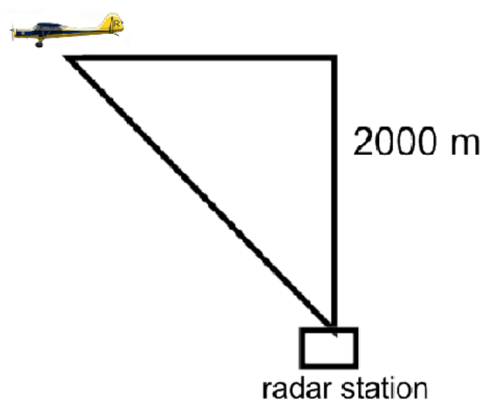
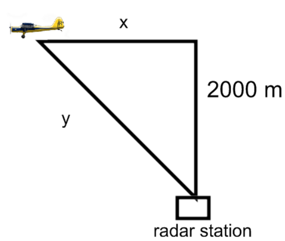
Problem 1 *A plane flying horizontally at an altitude of 2000 m and a speed of 200 m/sec passes directly over a radar station. (See figure below.) Find the rate at which the distance from the plane to the station is increasing 5 seconds after the plane passed directly over the radar station. Make sure to label the figure.*





Explanation.

$$\begin{aligned}
y^2 &= 2000^2 + x^2 \\
2y\frac{dy}{dt} &= 2x\frac{dx}{dt} \text{ (see } x \text{ and } y \text{ solved below)} \\
\sqrt{1000^2 + 2000^2}\left[\frac{dy}{dt}\right]_{t=5} &= 1000 \cdot 200 \\
\left[\frac{dy}{dt}\right]_{t=5} &= \frac{1000 \cdot 200}{\sqrt{1000^2 + 2000^2}} \text{ m/s}
\end{aligned}$$

Solving for x when $t = 5$:
We use the distance formula. Distance=rate*time, $d = rt$. Here we have $x = rt$.
$x = (5)(200) = 1000$ m

Solving for y when $t = 5$: